10.7 Gravity Inside a Shell

WORKSHEET

The question

STEP III 1999, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q9

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself Consider two small, diametrically opposite pieces of the ring. Draw the forces they exert on the particle. Which components cancel, and which add?

STAGE 2 Slicing the shell into rings

Ask yourself Put the centre of the shell at O and the particle on the axis at distance b from O. Slice the shell into thin rings perpendicular to the axis, labelled by the angle ϕ that a point of the ring makes with the axis at O. What are the radius, the position and the mass of the ring?

STAGE 3 The right variable

Ask yourself The denominator involves $s^2 = a^2 - 2ab\cos\phi + b^2$. What does s represent geometrically? What is $\mathrm{d}s$ in terms of $\mathrm{d}\phi$?

STAGE 4 It vanishes

Pause and try What changes if the particle is *outside* the shell, with $b > a$? The limits of s become $b - a$ and $b + a$. Redo the last line and show that the force is $\frac{GM}{b^2}$ towards the centre, as if all the mass were concentrated there.

Try this next

1 **Outside the shell.** Complete the Pause box: show that outside the shell the force is $\frac{GM}{b^2}$. Deduce that a solid uniform sphere of mass M pulls an outside particle as if its mass were at the centre.

2 **The collapse time.** Try STEP III 1987, Question 14 (step.maths.org/questions/87-s3-q14). It finds how long a cloud of dust takes to collapse under its own gravity; a trigonometric substitution does the integral.

3 **A tunnel through the Earth.** Inside a uniform solid sphere, show that the force at distance b from the centre is $\frac{GMb}{a^3}$, proportional to b. Deduce that a ball dropped down a tunnel through the centre of the Earth oscillates with period $2\pi\sqrt{a^3/(GM)}$, about 84 minutes, the same as the period of a satellite skimming the surface.

4 **Another force law.** If gravity followed an inverse-cube law, would the force inside a

shell still be zero? (Repeat Stage 3 with $\frac{1}{s^3}$ in place of $\frac{1}{s^2}$.)

Further reading

- **Daniel Kleppner & Robert Kolenkow, *An Introduction to Mechanics*** (Cambridge University Press), the chapters on gravitation, including the shell theorem.
- **Subrahmanyan Chandrasekhar, *Newton's Principia for the Common Reader*** (Oxford University Press). Newton's proofs, including Propositions 70 and 71, explained by a Nobel laureate.
- **Andrew Pontzen, *The Universe in a Box*** (Riverhead). How gravity, collapse and simulation explain the structure of the universe.

The full guided solution, and what the question leads to at university, is in Section 10.7:
https://beyondstep.keithmcnulty.org/ch10/shell-theorem.html